

When I ask a student to



Repeat what I just explained.

A Quick Recap



Q: Can the researcher manipulate variables in ex post facto research?

A: No, the researcher studies existing conditions without manipulation.

Q: Is it possible to establish cause and effect in ex post facto studies?

A: Not strongly, because the variables are not controlled.

Q: Can a cross-sectional study track changes over time?

A: Cross-sectional studies only provide a snapshot at one point in time

Q:Give one advantage of cross-sectional research.

A: It is quick and less expensive compared to longitudinal research.

A Quick Recap



Q: What is longitudinal research?

A: It studies the same subjects repeatedly over a period of time.

Q: Name one disadvantage of longitudinal research.

A: It takes a long time and may have participant dropout.

Q: What is one benefit of longitudinal studies?

A: They help observe changes and patterns over time.

Q: What does a correlational study measure?

A: The relationship between two or more variables.

Q: Does correlation mean causation?

A: No, just because two things are related doesn’t mean one causes the other.

A Quick Recap



Q: Correlation does not mean causation give example?

A: A study finds a correlation between the number of dental visits and the presence of dental caries. Does going to the dentist cause caries? No. People with more dental problems (like caries) are simply more likely to visit the dentist. The underlying cause is poor oral hygiene or diet, not the dental visits themselves.

Q:What is survey research?

A: It involves collecting data by asking people questions.

Q: What is one limitation of survey research?

A: People may not give honest answers.

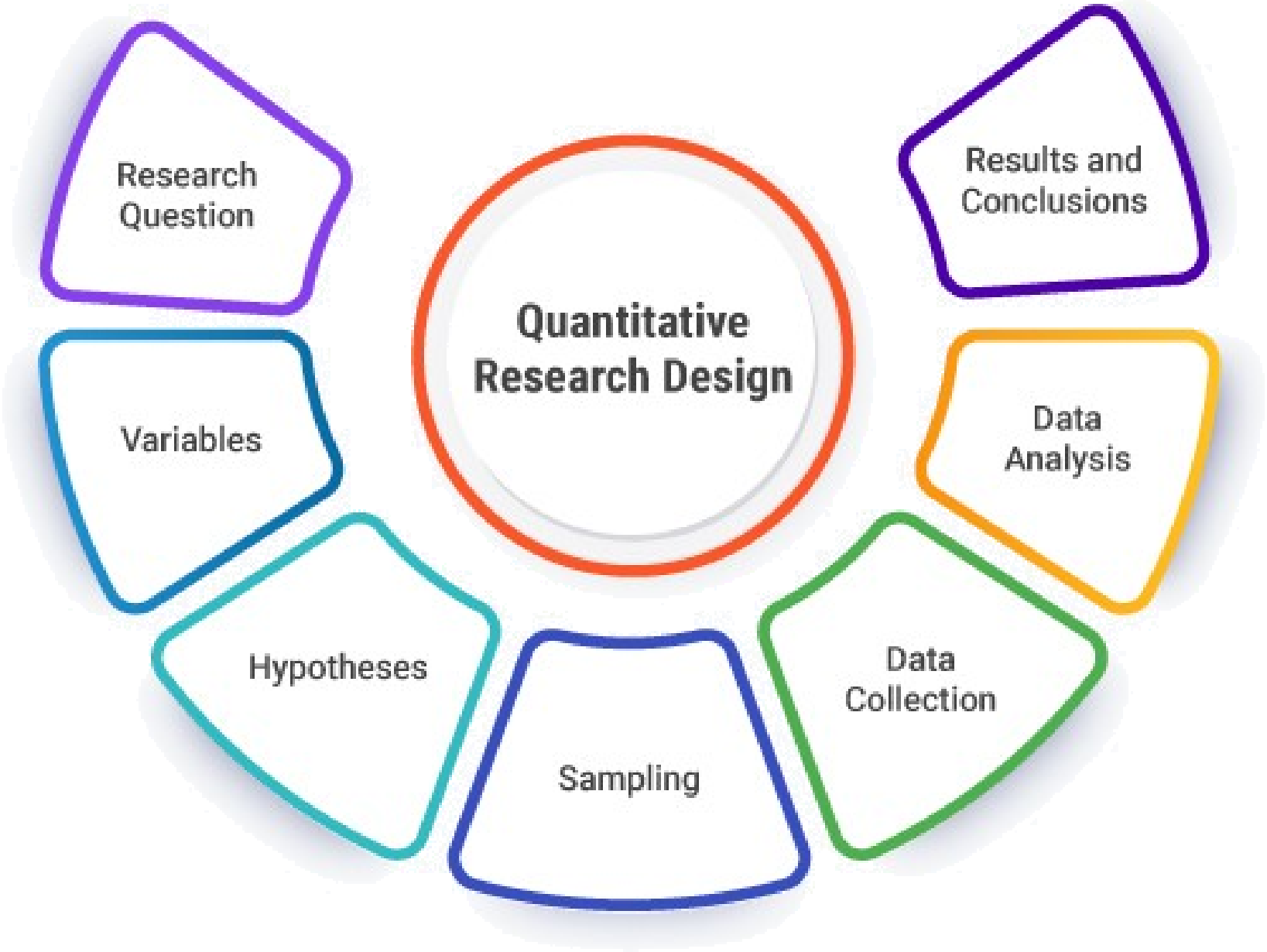
Objectives



At the end of the lecture student should be able to:

- Describe key components of a quantitative research study
- Explain basic data types (nominal, ordinal, interval, ratio) used in quantitative research.
- Understand the role of statistics in analyzing and interpreting quantitative data.
- Discuss ethical considerations related to conducting quantitative research involving human subjects.
- Differentiate between quantitative and qualitative research methods.

Quantitative Research: Components



Research question is a clear, focused question that guides a research study.

Variable is any characteristic that can take on different values, like height, temperature, or test scores.

Hypothesis is a testable statement that proposes a possible explanation for an observed phenomenon or relationship between variables

Sampling means selecting the group that you will actually collect data from in your research.

Data collection is the process of collecting and evaluating information or data from multiple sources to find answers to research problems.

Research data analysis is the systematic process of interpreting and extracting meaningful insights from data to answer research questions, test hypotheses, and draw conclusions.

Results are the presentation of data and hence findings or investigations. Conclusion is to write the output of the work/ investigations in summarized form.

Question: Does the use of interactive digital learning tools enhance pathology exam performance among dental students compared to traditional textbook-based learning?

Variables: Independent Variable-Type of learning

Dependent Variable-Exam performance

Hypothesis: Dental students who use interactive digital learning tools perform better in pathology exams than those who rely solely on traditional textbooks

Sampling: Second year BDS students, randomization

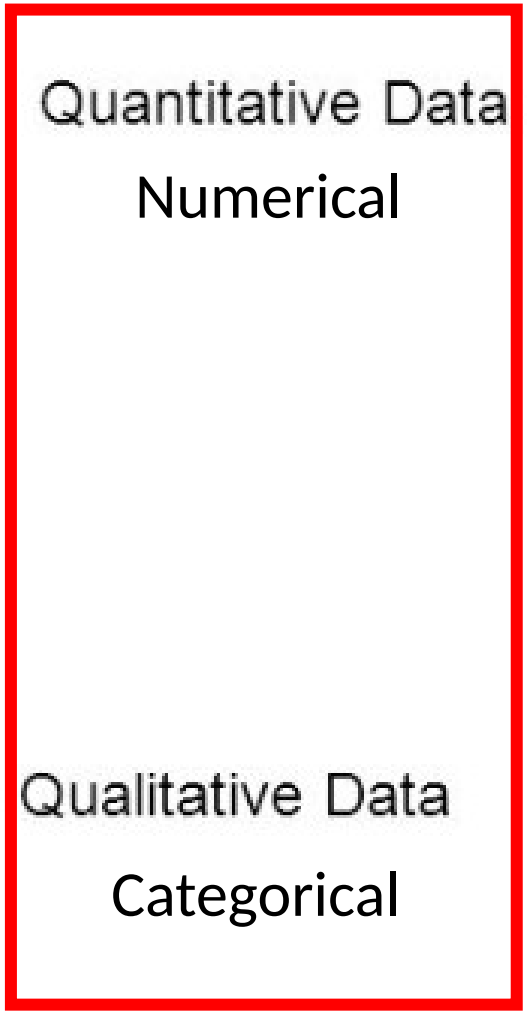
Data Collection:

Data Analysis:

Descriptive Statistics (Mean, median, standard deviation of exam scores in both groups)

Inferential Statistics (Independent t-test: To compare mean exam scores between groups)

Quantitative Research: Data Types



Quantitative Research: Data Types

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

Quantitative Research: Data Types



Quantitative Research: Role of Statistics



- Statistics???

- Statistics involves the collection, analysis, interpretation, and presentation of data. It provides methods for understanding and making inferences from numerical information, helping to summarize, describe, and make predictions about a population based on sample data.

Quantitative Research: Role of Statistics

- Summarizing Data: Descriptive statistics (mean, median, mode) help summarize patient data.

Example: If you collected the number of decayed teeth from 50 patients, statistics help you find the average number of decayed teeth.

- Comparing Groups: You can use statistics to compare two treatment methods.

Example: You want to know whether fluoride varnish or toothpaste is better in preventing caries. Statistics help you compare the results from two groups of patients.

- Finding Relationships: Correlation shows how two factors are related.

Example: Is there a relationship between sugar intake and number of cavities? Statistics can tell us how strong that relationship is.

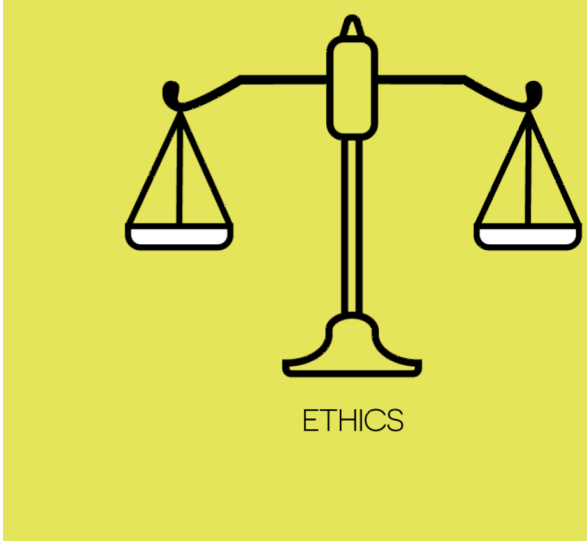
- Making Predictions: Statistics allows you to predict future trends based on data.

Example: Based on current data, you can estimate how many patients in your clinic might need dentures in the next 10 years.

- Checking Reliability: Hypothesis testing helps find out if a result is due to chance or is really meaningful.

Example: If a new mouthwash reduces plaque in 80% of users, statistics help you decide if this result is trustworthy or just a coincidence.

Quantitative Research: Role of Ethics



- Why Ethics Matter???

- In research involving people like patients or volunteers, we must always protect their rights, safety, and dignity. That’s what ethics is all about

Quantitative Research: Role of Ethics

- Informed Consent: Participants must know what the research is about, what they will do, and any risks involved.

Example: If you're surveying patients about oral hygiene habits, they must agree voluntarily, understanding it's part of a study.

- Confidentiality: Personal data must be kept private and secure.

Example: If a patient shares information about dental visits, their identity should not be revealed in your report.

- Voluntary Participation: No one should be forced or pressured to participate.

Example: Patients in a clinic should never feel like they must take part in a study to receive treatment.

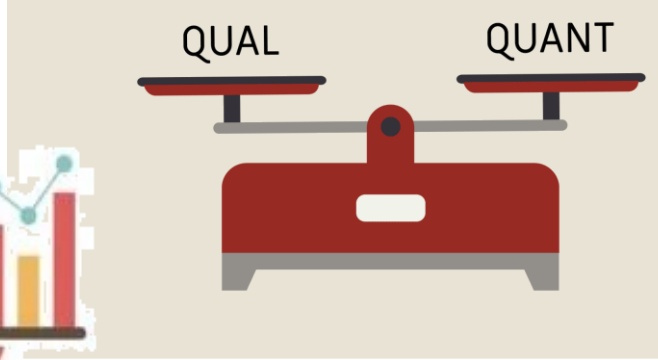
- No Harm to Participants: Research should not cause physical, psychological, or emotional harm.

Example: Don’t ask distressing or overly personal questions that may upset someone.

- Right to Withdraw: Participants can leave the study at any time, for any reason.

Example: Even if a patient signed up, they can choose to stop participating midway.

- Approval from Ethics Committee: All research involving humans must be reviewed by an Institutional Ethics Committee



Qualitative

Quantitative

- Purpose
- Data type
- Approach
- Analysis



- **Is there a correlation between daily sugar intake and the number of dental caries in children?**
- Type: Correlational Research(It examines the relationship between two naturally occurring variables without manipulation.)
- **What percentage of university students use mouthwash regularly?**
- Type: Cross-Sectional Research(Data is collected at a single point in time to assess the prevalence of a behavior.)
- **Does using a powered toothbrush reduce plaque levels more than a manual toothbrush over 8 weeks?**
- Type: Experimental Research(Involves manipulation of variables and random assignment to groups.)

- **Are adolescents who consumed carbonated drinks in early childhood more likely to develop enamel erosion?**
- Type: Ex Post Facto Research(Looks backward to study possible causes after the effect has occurred, without manipulation.)
- **How does oral hygiene behavior change in orthodontic patients over a 12-month period?**
- Type: Longitudinal Research(Tracks changes in variables over time in the same group of individuals.)
- **Is there a relationship between screen time and academic performance among students?**
- Type: Correlational Research(Explores the association between two variables without manipulation.)

- **What is the average number of hours university students sleep during exam week?**
- Type: Descriptive (Cross-Sectional) Research(Collects data at one point in time to describe a population characteristic.)
- **Does a structured exercise program improve blood pressure in adults over 50?**
- Type: Experimental Research(Involves intervention and comparison of outcomes between groups.)
- **Do people who experienced job loss during the pandemic show higher rates of anxiety one year later?**
- Type: Ex Post Facto Research(Studies the effect of an existing condition or event retrospectively.)

- **How does students' stress level change from the beginning to the end of the academic year?**
- Type: Longitudinal Research (Measures change in the same subjects over an extended period.)
- **A group of researchers wants to find out whether students who attend online lectures perform better in exams than those who attend face-to-face lectures. Which part of the research process is described in the above statement?**
- Answer: Research Question
- **In a study measuring the effect of sleep duration on academic performance among first-year medical students. What are the independent and dependent variables?**
- Sleep duration (independent), Sleep duration (dependent)

- **The number of hours students study per week was recorded in a research study. What type of variable is “hours of study”? (Nominal, Ordinal, Interval, or Ratio)**
- Answer: Ratio
- **Researchers used a questionnaire to gather data on student motivation before and after a gamified learning intervention. Which step of research does this represent?**
- Answer: Data Collection
- **A researcher predicts that students who use spaced repetition techniques will score higher on the final exam than those who use passive re-reading. Which step of research is shown here?**
- Answer: Hypothesis

- **In a study to examine the effect of physical exercise on stress levels among students. What is the dependent variable?**
- Answer: Stress levels
- **The severity of a disease is rated as mild, moderate, or severe. What type of data is this?**
- Answer: Ordinal (The categories have a meaningful order but the difference between them is not exactly measurable)
- **A research study records the time (in minutes) each student spends on a pathology quiz. What type of data is “time in minutes”?**
- Answer: Ratio (have a true zero and equal intervals between values)

- **Which of the following is an example of nominal data?**

A. Blood type (A, B, AB, O)
B. Pain score from 1 to 10
C. Temperature in Celsius
D. Height in centimeters

A:Blood type is categorized into distinct groups without order

- **Which of the following is an example of interval data?**

A. Weight in kilograms
B. Number of siblings
C. Temperature in Celsius
D. Stages of cancer (I, II, III, IV)

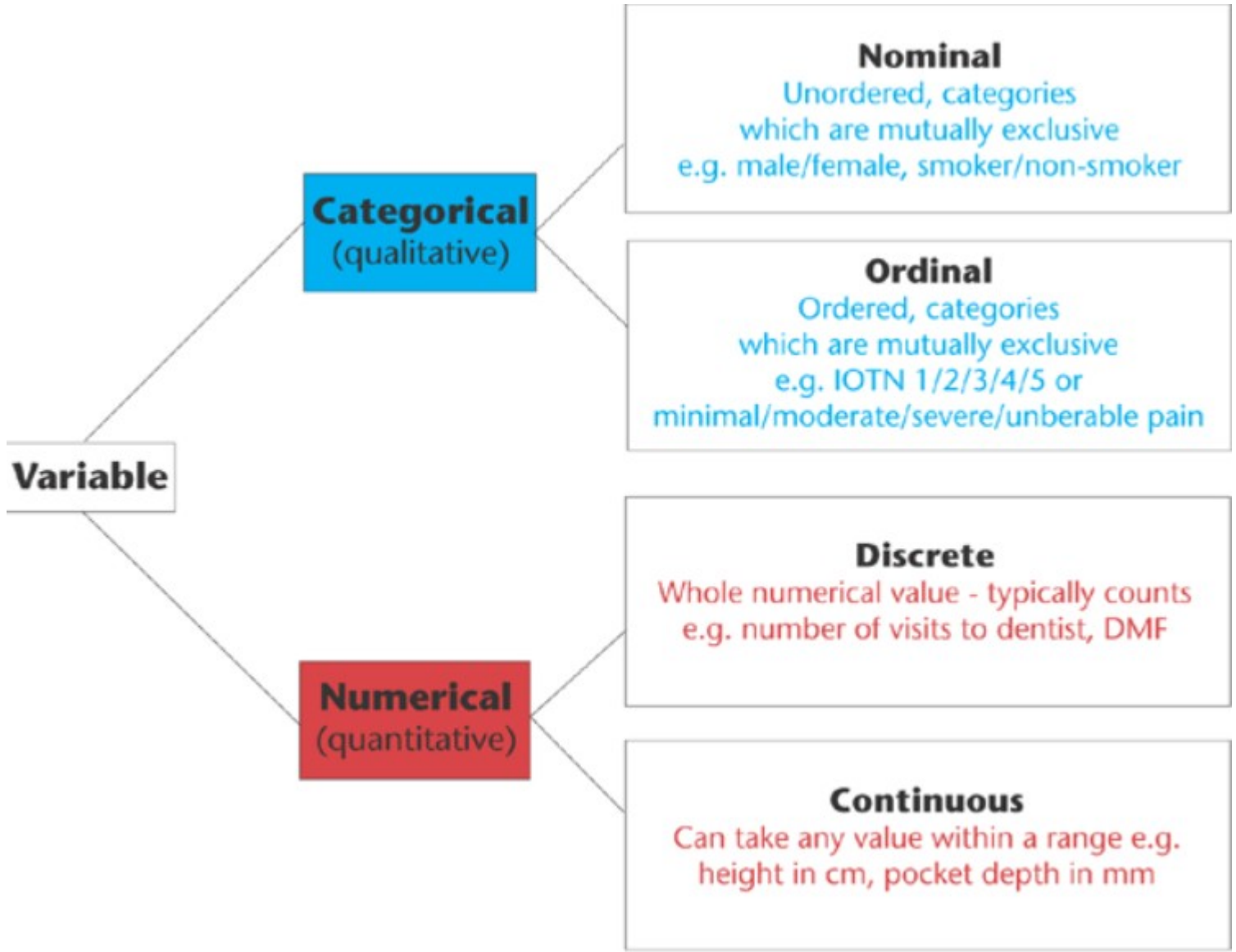
C: Interval data have equal differences between values but no true zero.

- **In the research process, which step involves determining whether the difference in test scores between two groups is statistically significant?**

A. Data Collection
B. Sampling
C. Hypothesis
D. Data Analysis

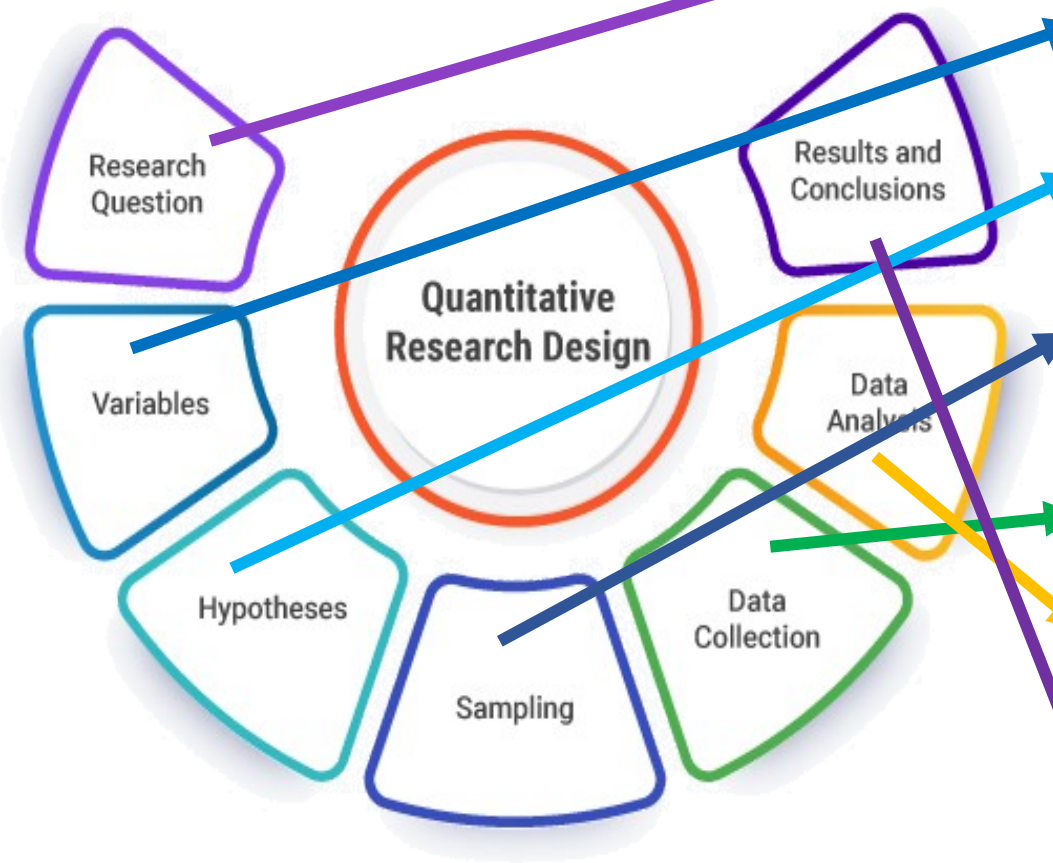
D: Data analysis involves interpreting data to test hypotheses and draw conclusions, including statistical testing.


Not sure if thank you for your attention
or thank you for not sleeping during the presentation


Variable
Categorical
(qualitative)
Nominal
Unordered, categories
which are mutually exclusive
e.g. male/female, smoker/non-smoker
Ordinal
Ordered, categories
which are mutually exclusive
e.g. IOTN 1/2/3/4/5 or
minimal/moderate/severe/unberable pain
Numerical
(quantitative)
Discrete
Whole numerical value - typically counts
e.g. number of visits to dentist, DMF
Continuous
Can take any value within a range e.g.
height in cm, pocket depth in mm

- Hypothesis:Final-year dental students experience higher levels of academic stress compared to first-year dental students.
- Hypothesis:Regular simulation lab practice improves cavity preparation accuracy among second-year dental students.
- Hypothesis:Dental students in their clinical years have a more positive attitude toward treating patients with special needs than those in preclinical years.
- Hypothesis:Dental students who follow a structured daily schedule report higher academic performance than those who do not.
- Hypothesis: The use of mobile apps for dental anatomy learning increases retention of anatomical knowledge among first-year dental students.

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